
One Short of Steinitz: Weighted Regression Selection

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Abstract

How well can least squares perform on a full dataset after training on a small weighted subset? A recent COLT open problem determined the exact worst-case ratio for budgets at most the dimension and at least twice the dimension, leaving every intermediate budget open and reporting an unpublished endpoint argument. We give a complete proof that, for d -dimensional homogeneous linear regression and the minimum-norm ERM, the exact ratio with $2d - 1$ selected examples is $1 + 1/d$.

The proof identifies the only obstruction to exact recovery one example below the Steinitz threshold. Unless the nonzero full-loss gradients lie on d independent lines and every useful zero-gradient feature is absent, $2d - 1$ examples recover the full ERM exactly. On the exceptional cross configuration, the problem splits into one-dimensional regressions and omitting one member of the least-loss line costs at most a $1/d$ fraction of the optimum. A matching construction handles the minimum-norm tie rule.

We also give a harmonic lower bound for every intermediate budget and show that it is exact for rank-additive circuit decompositions. This class contains the extremal construction and isolates the remaining obstacle for the full open curve: overlapping positive circuits can only be handled by a new reduction, not by ordinary volume sampling.

1 Introduction

Data selection asks a deliberately data-centric question. Fix the training algorithm, reveal the entire finite dataset, and choose a small subset on which to train. How close can the resulting model come to the model trained on all examples? This differs from feature selection, random subsampling, and label-efficient

active learning. The selector here knows both features and responses, may assign nonnegative weights, and is evaluated on the original observed dataset.

Even homogeneous least squares has a sharp unresolved regime. Let $\mathcal{A}_*$ return the minimum Euclidean-norm empirical risk minimizer. For a dataset in $\mathbb{R}^d \times \mathbb{R}$, Hanneke et al. (2025a) proved that the worst multiplicative loss ratio is infinite below d selected examples, equals $d + 1$ at d , and equals one at $2d$ or more. Their COLT open-problem note asks for the exact value between d and $2d$. The authors report belief in an unpublished, ad hoc argument for the value $1 + 1/d$ at $2d - 1$ (Hanneke et al., 2025b). The two proved endpoints use different geometric tools. Size- d volume sampling gives $d + 1$ (Dereziński and Warmuth, 2017, 2018a), while Steinitz’s theorem gives exact recovery at $2d$ (Steinitz, 1916).

We give a complete endpoint proof. The argument does not interpolate an existing volume-sampling estimate. Instead, it characterizes when the $2d$ -point Steinitz guarantee cannot be improved by one point. In that case all informative gradients form a cross, namely d pairs of opposite directions. The regression objective then separates over those directions. A one-dimensional averaging argument chooses which pair to leave incomplete and gives the factor $1 + 1/d$.

Our contributions are:

- We prove the exact universal identity

$$\sup_D \frac{L_D^*(2d - 1; \mathcal{A}_*)}{L_D^*} = 1 + \frac{1}{d}. \quad (1)$$

This includes rank-deficient datasets, zero optimal loss, arbitrary dataset size, and the minimum-norm tie rule.

- We characterize the datasets whose full optimum needs $2d$ weighted examples to become a *unique* selected ERM. Their nonzero gradients lie on d independent lines, both orientations occur on each line, and every zero-gradient feature is zero. We separately state the exact criterion for the

minimum-norm rule, which can exploit a rank-deficient objective.

- For every budget $d + k$, we give an explicit harmonic lower bound. We prove the same expression as an upper bound whenever the normalized loss admits a rank-additive conformal circuit decomposition. Direct sums of balanced simplex blocks are extremal in this class.

The last result is deliberately not presented as a solution of the full intermediate curve. Positive-basis structure permits overlapping circuits (Davis, 1954; Schoch, 2020; Hare et al., 2023). Our exact argument covers rank-additive circuits, and computations suggest overlap helps, but a universal reduction remains open.

Relation to other subset problems. Strong regression coresets preserve the objective uniformly over all parameters, which is a different and generally more demanding target (Boutsidis et al., 2013; Feldman, 2020). Active regression chooses labels without seeing all responses (Chen and Price, 2019). Machine teaching also optimizes training examples for a fixed learner, but commonly allows synthesis or targets parameter identification rather than the full-dataset loss (Ma et al., 2018). The present formulation follows Hanneke et al. (2025a): labels are known, selected examples must come from the dataset, and the learner is fixed.

2 Problem and Geometry

Let $D = \{(x_i, y_i)\}_{i=1}^N$ be a finite multiset in $\mathbb{R}^d \times \mathbb{R}$. Define

$$L_D(w) = \frac{1}{N} \sum_{i=1}^N (x_i^\top w - y_i)^2, \quad L_D^* = \min_w L_D(w). \quad (2)$$

For a convex combination $F = \sum_i \lambda_i \ell_i$, where $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$, and $\ell_i(w) = (x_i^\top w - y_i)^2$, let $\mathcal{A}_*(F)$ be its minimum-norm minimizer. The best loss using at most n examples is

$$L_D^*(n; \mathcal{A}_*) = \inf_{\substack{\lambda \geq 0 \\ |\text{supp}(\lambda)| \leq n}} L_D\left(\mathcal{A}_*\left(\sum_i \lambda_i \ell_i\right)\right). \quad (3)$$

Repeated examples and zero weights make this equivalent to the original definition with n selected examples. Ratios use the conventions of Hanneke et al. (2025a): $0/0 = 1$ and a positive numerator over zero is infinite. Assume first that the feature matrix has rank d . The

full minimizer w_* is unique. Put

$$r_i = y_i - x_i^\top w_*, \quad g_i = r_i x_i, \quad H = \frac{1}{N} \sum_i x_i x_i^\top. \quad (4)$$

The normal equations and Pythagorean identity are

$$\sum_i g_i = 0, \quad L_D(w_* + v) = L_D^* + v^\top H v. \quad (5)$$

The vectors g_i are the individual gradients at w_* up to a common nonzero scalar and sign.

The tie rule enters through one elementary criterion.

Lemma 1 (Exact selected ERM). *Let S be the positive-weight support of a selected objective and let $W_S = \text{span}\{x_i : i \in S\}$. Its minimum-norm ERM equals w_* if and only if*

$$\sum_{i \in S} \lambda_i g_i = 0 \quad \text{and} \quad w_* \in W_S. \quad (6)$$

If $W_S = \mathbb{R}^d$, the second condition is automatic and w_* is the unique selected ERM.

The first condition says that w_* minimizes the selected quadratic. Its minimizer set is then $w_* + W_S^\perp$. The closest member to zero is w_* exactly when $w_* \perp W_S^\perp$, proving the lemma. This distinction is important. Translating w_* to zero preserves loss differences but not the minimum-norm choice on rank-deficient supports.

Positive bases. A finite set positively spans a vector space if every vector is a nonnegative combination of its elements. An inclusion-minimal positive spanning set is a positive basis. A positive basis of $\mathbb{R}^q$ has between $q+1$ and $2q$ vectors. At the upper endpoint it is a cross: after rescaling, it has the form $\{\pm u_1, \dots, \pm u_q\}$ for a linear basis $u_1, \dots, u_q$ (Davis, 1954; Regis, 2016).

3 One Example Below Exactness

We first isolate the only configuration in which the standard $2d$ exact certificate genuinely needs all $2d$ examples.

Definition 2 (Unique-certificate number). *For a full-rank dataset, let $c_u(D)$ be the least support size of a weighted objective for which w_* is the unique minimizer.*

Theorem 3 (Cross dichotomy). *Every full-rank dataset satisfies $c_u(D) \leq 2d$. Equality holds if and only if there are independent lines $U_1, \dots, U_d$ such that*

1. every nonzero g_i lies in some U_j ,
2. both orientations occur among the g_i on every U_j , and

3. $g_i = 0$ implies $x_i = 0$.

If these conditions fail, $c_u(D) \leq 2d - 1$.

Proof. Let $V = \text{span}\{g_i : g_i \neq 0\}$ and $q = \dim V$. The nonzero g_i have a strictly positive combination equal to zero. Steinitz's theorem therefore selects at most $2q$ of them that positively span V . Every nonzero-gradient feature belongs to V . Since all features span $\mathbb{R}^d$, we may add $d - q$ zero-gradient features that extend V to $\mathbb{R}^d$. Their union gives a unique exact objective on at most $d + q$ examples. Thus $q < d$ gives at most $2d - 1$.

Now let $q = d$ and choose a positive basis B from the nonzero gradients. If $|B| \leq 2d - 1$, its positive dependence and full span give the claim. Otherwise B is a cross $\{\pm u_j\}_{j=1}^d$. Write any other nonzero gradient as

$$g = \sum_{j \in J} a_j s_j u_j, \quad a_j > 0, \quad s_j \in \{-1, 1\}. \quad (7)$$

If $|J| \geq 2$, then

$$\{g\} \cup \{-s_j u_j : j \in J\} \cup \{\pm u_\ell : \ell \notin J\} \quad (8)$$

positively spans $\mathbb{R}^d$ and has $2d - |J| + 1 \leq 2d - 1$ elements. Hence every nonzero gradient must lie on a cross line if $c_u(D) = 2d$.

Let Z be the span of zero-gradient features and put $z = \dim Z$. If $z > 0$, choose z such features and $d - z$ cross lines that together span $\mathbb{R}^d$. A balancing pair from every chosen line plus the zero-gradient features gives a unique exact objective on $2(d - z) + z = 2d - z$ examples. Thus $z = 0$ is necessary.

Conversely, under the three stated conditions, full selected feature rank requires using every line. Gradient balance holds independently on each line and needs both orientations. Therefore at least $2d$ positive-weight examples are necessary, while one balancing pair per line is sufficient. $\square$

The theorem concerns uniqueness. By Lemma 1, a rank-deficient objective can still return w_* when its feature span contains w_* . On the exceptional branch, this tie behavior also has an exact description.

Corollary 4 (Min-norm compression on a cross). *Suppose the three conditions of Theorem 3 hold, and write $w_* = \sum_{j=1}^d \beta_j u_j$. The least support size of a weighted objective whose minimum-norm ERM is w_* equals one if $w_* = 0$ and the dataset contains a zero-feature example. In every other case it equals*

$$2 \max\{1, |\{j : \beta_j \neq 0\}|\}. \quad (9)$$

For every nonzero coordinate β_j , the selected feature span must include U_j . Gradient balance on an independent cross line requires both orientations. Conversely, balancing pairs on precisely those lines have a minimizer set whose feature span contains w_* . If $w_* = 0$, any one balancing pair suffices, while a zero-feature loss has minimum-norm ERM zero by itself. Thus $2d$ is necessary for the raw min-norm rule exactly when every β_j is nonzero.

Theorem 5 (Sharp endpoint). *For every integer $d \geq 1$,*

$$\sup_{D \subset \mathbb{R}^d \times \mathbb{R}} \frac{L_D^*(2d - 1; \mathcal{A}_*)}{L_D^*} = 1 + \frac{1}{d}. \quad (10)$$

Upper bound. If the feature rank is $r < d$, the known $2r$ exact guarantee fits within $2d - 1$. The same is immediate when $L_D^* = 0$. Assume full rank and positive optimal loss. If Theorem 3 gives a unique exact support of size at most $2d - 1$, there is nothing to prove. We may therefore work under its three cross conditions.

Choose a nonzero vector u_j on each cross line and write $x_i = \alpha_i u_j$ for indices $i \in I_j$. Put

$$h_j = \frac{1}{N} \sum_{i \in I_j} \alpha_i^2, \quad A_j = \frac{1}{N} \sum_{i \in I_j} r_i^2. \quad (11)$$

Examples with $x_i = 0$ may add a constant to L_D^* but do not enter these sums. Let $q_1, \dots, q_d$ be the dual basis, so $u_\ell^\top q_j = \mathbf{1}\{\ell = j\}$. Equation (5) gives

$$(tq_j)^\top H(tq_j) = h_j t^2. \quad (12)$$

Moreover A_j/h_j is the α_i^2 -weighted average of $(r_i/\alpha_i)^2$. Hence there is an $i(j) \in I_j$ with

$$h_j \left(\frac{r_{i(j)}}{\alpha_{i(j)}} \right)^2 \leq A_j. \quad (13)$$

The line normal equation $\sum_{i \in I_j} r_i \alpha_i = 0$ provides an example with opposite gradient orientation to $i(j)$.

Pick a line j with $A_j \leq L_D^*/d$. On every other line choose a balancing pair. On line j keep only $i(j)$. Weight each pair so its gradient at w_* cancels. The selected features span $\mathbb{R}^d$, and the selected objective separates in the coordinates $u_\ell^\top (w - w_*)$. Its unique minimizer is

$$w_* + \frac{r_{i(j)}}{\alpha_{i(j)}} q_j. \quad (14)$$

Equations (12) and (13) bound its full loss by $L_D^* + A_j \leq (1 + 1/d)L_D^*$. $\square$

Lower bound. Use the $2d$ examples

$$(e_j, 0), \quad (-e_j, -2), \quad j = 1, \dots, d. \quad (15)$$

The full ERM is $w_* = (1, \dots, 1)$ and $L_D^* = 1$. Any support of size at most $2d - 1$ leaves some coordinate with at most one selected example. A singleton makes that coordinate zero or two. With no selected example, the minimum-norm rule makes it zero. Its two full-data losses therefore sum to four instead of the optimal two. All other coordinates contribute at least their optimal value, so the ratio is at least $1 + 1/d$. Equal weights on complete pairs and one singleton attain equality. $\square$

Proposition 6 (Instance-dependent cross bound). *Under the cross conditions, define*

$$B_j = h_j \min_{i \in I_j} \left(\frac{r_i}{\alpha_i} \right)^2. \quad (16)$$

There is a full-rank support of size $2d - 1$ whose full-loss excess is exactly $\min_j B_j$. Moreover,

$$\min_j B_j \leq \min_j A_j \leq \frac{L_D^*}{d}. \quad (17)$$

The construction in the upper-bound proof gives the first statement after choosing the best line and singleton. The averaging argument (13) gives the inequalities. Thus a line containing an example with a small residual-to-feature ratio can make the endpoint loss much smaller than the universal $1 + 1/d$ bound.

Generic datasets are easier. If the nonzero gradients span $\mathbb{R}^d$ and are in linear general position, an inclusion-minimal positive dependence has exactly $d + 1$ elements and spans $\mathbb{R}^d$. Thus $d + 1$ examples recover w_* uniquely. The intermediate worst case is entirely a degeneracy phenomenon, with the endpoint extremum at the maximally degenerate cross.

Constructive extraction. All supports in the proof can be found in polynomial time. Compute w_* , the gradients g_i , and bases for their span and the zero-gradient feature span. Repeatedly delete a gradient whenever the remainder still positively spans the same space. Each test is a linear program, and the terminal set is a positive basis of size at most twice its rank. A size- $2d$ terminal basis is a cross. Equation (8) either replaces it by a smaller exact certificate, or (16) chooses the singleton. Positive balancing weights follow from one final linear feasibility problem. The endpoint guarantee is therefore constructive, not only existential.

4 All-Budget Harmonic Lower Bound

Write the intermediate budget as $n = d + k$ with $0 \leq k < d$. For $m \in \{k + 1, \dots, d\}$ define

$$H(d, m) = \min_{\substack{b_j \in \mathbb{N} \\ \sum_{j=1}^m b_j = d}} \sum_{j=1}^m \frac{1}{b_j}. \quad (18)$$

Table 1: The lower-bound excess $\Psi(d, k)$ for small dimensions. The selected budget is $d + k$.

$d \backslash k$	0	1	2	3	4	5
2	2	1/2				
3	3	2/3	1/3			
4	4	1	1/2	1/4		
5	5	6/5	3/5	2/5	1/5	
6	6	3/2	2/3	1/2	1/3	1/6

If $d = qm + r$ with $0 \leq r < m$, convexity of $1/x$ gives

$$H(d, m) = \frac{m - r}{q} + \frac{r}{q + 1}. \quad (19)$$

Set

$$\Psi(d, k) = \max_{m=k+1, \dots, d} \frac{m - k}{H(d, m)}. \quad (20)$$

Theorem 7 (Intermediate lower bound). *For every $0 \leq k < d$,*

$$\sup_D \frac{L_D^*(d + k; \mathcal{A}_*)}{L_D^*} \geq 1 + \Psi(d, k). \quad (21)$$

At $k = 0$ this is $d + 1$, and at $k = d - 1$ it is $1 + 1/d$.

The construction uses orthogonal simplex blocks. For a composition $d = b_1 + \dots + b_m$, block j contains the standard $b_j + 1$ -point simplex lower instance for size- b_j regression. Give it optimal-loss share

$$a_j = \frac{1/b_j}{\sum_{\ell=1}^m 1/b_\ell}. \quad (22)$$

Using b_j examples in that block incurs excess $b_j a_j$, while using all $b_j + 1$ removes the excess. A large parameter offset makes fewer than b_j examples no better under the minimum-norm rule. Every finite-ratio selection spends the baseline $\sum_j b_j = d$ examples, and each of the k remaining examples can complete one block. Equation (22) equalizes all block excesses, leaving $(m - k)/\sum_j 1/b_j$. The appendix gives the finite-offset construction and limiting argument.

The lower bound has two regimes up to integer effects. The maximizing m is near $2k$ while $k \lesssim d/2$, giving excess about $d/(4k)$. Once $m = d$ binds, the excess is $(d - k)/d$. The latter includes the endpoint proved universally above.

5 When the Harmonic Curve Is Exact

The lower construction belongs to a broader class on which (21) is also an upper bound. This result explains why simplex blocks arise and states precisely which reduction is missing in the general case.

Whiten the full loss around w_* so that

$$\sum_i p_i x_i x_i^\top = I, \quad \sum_i p_i r_i x_i = 0, \quad \sum_i p_i r_i^2 = 1. \quad (23)$$

Then the full loss ratio at $w_* + v$ is $1 + \|v\|^2$. Splitting an example weight into nonnegative pieces does not change the objective and lets one treat repeated circuit occurrences separately.

Definition 8 (Rank-additive circuit decomposition). *A normalized instance is rank-additive circuit decomposable if its weighted loss can be split into m centered circuit quadratics*

$$F_j(v) = \|U_j v - s_j\|^2 = a_j + v^\top H_j v, \quad H_j = U_j^\top U_j, \quad (24)$$

where U_j has $b_j + 1$ rows and rank b_j , $U_j^\top s_j = 0$, $a_j = \|s_j\|^2$, and

$$\sum_j H_j = I, \quad \sum_j a_j = 1, \quad \sum_j b_j = d. \quad (25)$$

Every row occurrence must be a nonnegative scaling of an original example loss.

The condition arises from a conformal circuit decomposition of the full residual in the kernel of the design. Rank additivity is the substantive part. General decompositions can have overlapping circuits whose ranks sum to more than d .

Theorem 9 (Circuit sparsification). *Under Definition 8, for every $K \subseteq [m]$ there is a weighted support of at most $d + |K|$ original examples with full-loss excess at most*

$$\sum_{j \notin K} b_j a_j. \quad (26)$$

Consequently the worst ratio over rank-additive circuit decomposable instances at budget $d + k$ is exactly $1 + \Psi(d, k)$.

Proof. Choose full-column factors $H_j = B_j B_j^\top$, with $B_j \in \mathbb{R}^{d \times b_j}$, and concatenate them as $B = [B_1 \cdots B_m]$. By (25), B is square and $BB^\top = I$. Thus B is orthogonal. It follows that the columns within each B_j are orthonormal, columns from different factors are orthogonal, and $H_j = B_j B_j^\top$ is the orthogonal projector onto its range.

Write $\tilde{U}_j = U_j B_j$. Since U_j vanishes on the orthogonal complement of $\text{range}(H_j)$, this gives the intrinsic factorization $U_j = \tilde{U}_j B_j^\top$. Moreover, $\tilde{U}_j^\top \tilde{U}_j = I_{b_j}$, $\tilde{U}_j^\top s_j = 0$, and $\|s_j\|^2 = a_j$. Hence the square augmented matrix

$$\begin{bmatrix} \tilde{U}_j & s_j/\sqrt{a_j} \end{bmatrix} \quad (27)$$

is orthogonal when $a_j > 0$.

Omit one of the $b_j + 1$ circuit rows with probability proportional to the squared volume of the remaining b_j design rows. The resulting interpolator v_j has

$$\mathbb{E} v_j = 0, \quad \mathbb{E}[v_j v_j^\top] = a_j H_j, \quad \mathbb{E}\|v_j\|^2 = b_j a_j. \quad (28)$$

These identities follow directly by viewing the omitted row of the orthogonal augmented matrix as its cofactor vector. They also follow from the size- b_j volume-sampling identity.

For $j \in K$, retain all $b_j + 1$ rows and use its centered circuit weights. For $j \notin K$, retain the random b_j -row basis. Orthogonality makes the global selected ERM the sum of the incomplete-circuit interpolators. Thus its expected squared norm is the right side of (26). Some deterministic choice is no worse. It uses $\sum_j b_j + |K| = d + |K|$ row occurrences and no more original examples.

If $m \leq k$, the selector completes every circuit and is exact. Otherwise, for fixed b_j , it completes the k largest values $b_j a_j$. The exact allocation identity for $m \geq k + 1$ is

$$\sup_{a \geq 0, \sum a_j \leq 1} \min_{|K|=k} \sum_{j \notin K} b_j a_j = \max_{\substack{J \subseteq [m] \\ |J| \geq k+1}} \frac{|J| - k}{\sum_{j \in J} 1/b_j}. \quad (29)$$

It is proved in the appendix. The active subset J allows zero loss on some circuit blocks. Optimizing also over the number and integer ranks of the circuits reduces the right side to $\Psi(d, k)$. Theorem 7 supplies equality. $\square$

6 Discussion

The endpoint $2d - 1$ is governed by a sharp geometric dichotomy. A single gradient involving two cross coordinates creates a smaller positive basis. A single nonzero zero-gradient feature also lowers the exact unique-ERM certificate. Only the cross survives, and its loss allocation forces the $1/d$ constant. This is why the endpoint admits an exact argument despite the mismatch between volume sampling and Steinitz sparsification.

The all-budget lower bound suggests the curve $1 + \Psi(d, k)$. Theorem 9 proves it whenever conformal circuits are rank additive. Positive-basis decompositions need not have this property. Their simplices may overlap through critical vectors (Schoch, 2020; Hare et al., 2023). Such overlap gives additional cancellation routes in small-dimensional exact optimization, but turning that observation into a universal inequality remains the central open step.

Regularized volume sampling controls inverse Hessians under stochastic noise assumptions (Dereziński

and Warmuth, 2018b). It does not control the fixed residual-induced bias here. Ordinary size- $s > d$ volume sampling also lacks a response-uniform loss identity (Dereziński and Warmuth, 2018a). The circuit proof succeeds because each local residual space is one-dimensional and rank additivity makes its covariance orthogonal to every other circuit.

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A Additional Linear-Algebra Details

Expanded proof of Lemma 1. The selected objective has gradient at w_* equal to $-2 \sum_{i \in S} \lambda_i g_i$. Therefore its first condition for optimality is exactly the first part of (6). Under this condition its value at $w_* + v$ differs from its value at w_* by

$$\sum_{i \in S} \lambda_i (x_i^\top v)^2. \quad (30)$$

Its minimizer set is $w_* + W_S^\perp$. The minimum-norm member of this affine space is the orthogonal projection of w_* onto W_S . It equals w_* if and only if $w_* \in W_S$. $\square$

Lemma 10 (Cross replacement). *Let $B = \{\pm u_1, \dots, \pm u_d\}$ be a cross. If a nonzero vector g has at least two nonzero coordinates in the basis (u_j) , then $B \cup \{g\}$ contains a positive spanning subset of size at most $2d - 1$.*

Proof. Write $g = \sum_{j \in J} a_j s_j u_j$ with $a_j > 0$ and $s_j \in \{-1, 1\}$. The set $\{g\} \cup \{-s_j u_j : j \in J\}$ positively spans the coordinate subspace indexed by J , because $g + \sum_{j \in J} a_j (-s_j u_j) = 0$ is a strictly positive dependence and the set spans that subspace. Add both orientations on every coordinate outside J . The union positively spans $\mathbb{R}^d$ and has $1 + |J| + 2(d - |J|) = 2d - |J| + 1 \leq 2d - 1$ elements. $\square$

B Finite Lower-Bound Construction

Fix a block dimension b and an offset $M > 0$. Use features

$$e_1, \dots, e_b, -\mathbf{1}_b \quad (31)$$

and responses

$$M, 2M, \dots, bM, -M \frac{b(b+1)}{2} - b - 1. \quad (32)$$

The full minimizer is

$$w_* = (M + 1, 2M + 1, \dots, bM + 1), \quad (33)$$

and every one of the $b+1$ residuals has magnitude one. The average optimal loss is one. Omitting any one point and interpolating the other b makes the omitted residual have magnitude $b+1$. Hence the full average loss is $b+1$, with excess b .

Consider any support with fewer than b distinct examples. Its selected rows are linearly independent, so every positive weighting has zero selected loss and returns the minimum-norm interpolator of those rows. If $-\mathbf{1}_b$ is absent, at least one omitted basis coordinate is set to zero, while its response has magnitude at least M . The full loss therefore diverges with M .

If $-\mathbf{1}_b$ is present, at most $b-2$ basis rows are present. Let J be the set of at least two omitted basis coordinates. The minimum-norm solution assigns a common value to every coordinate in J , because their only constraint is their sum. Two distinct targets in J differ by at least M . For any common prediction c and two such indices $q \neq r$,

$$(c - qM)^2 + (c - rM)^2 \geq \frac{(q - r)^2 M^2}{2} \geq \frac{M^2}{2}. \quad (34)$$

Thus the full loss again diverges as $M \rightarrow \infty$. This is the same offset device used for the sharp size- d regression lower bound of Hanneke et al. (2025a).

Take orthogonal copies with dimensions $b_1, \dots, b_m$. Scale all responses in block j so its contribution to the full optimal loss is a_j . Orthogonal features make both the selected ERM and full loss decompose block-wise. In the limit $M \rightarrow \infty$, a finite-ratio support uses at least b_j examples in every block. With exactly b_j its excess contribution is $b_j a_j$. With $b_j + 1$ it is zero. A budget $d + k$ completes at most k blocks. Choosing (22) proves Theorem 7. For every finite M this is a finite dataset, so taking $M \rightarrow \infty$ is valid for the supremum in (21).

C The Allocation Identity

Proof of (29). Put $c_j = b_j a_j$ and let $T(c)$ be the sum of the $m - k$ smallest c_j . For any active set J with $|J| \geq k + 1$, set $c_j = t$ on J and zero outside, where $t = (\sum_{j \in J} 1/b_j)^{-1}$. This is feasible and gives $T(c) = (|J| - k)t$, proving the lower bound in (29).

For the reverse bound use the layer-cake identities

$$T(c) = \int_0^\infty (|J_t| - k)_+ dt, \quad (35)$$

$$\sum_j \frac{c_j}{b_j} = \int_0^\infty \sum_{j \in J_t} \frac{1}{b_j} dt, \quad (36)$$

where $J_t = \{j : c_j \geq t\}$. If the right side of (29) is Φ , its definition gives $(|J_t| - k)_+ \leq \Phi \sum_{j \in J_t} 1/b_j$ at every t . Integration gives $T(c) \leq \Phi \sum_j c_j/b_j \leq \Phi$. $\square$

To optimize over compositions, fix an active subset J of size ℓ . Since $\sum_{j \in J} b_j \leq d$, adding unused dimensions to these blocks can only decrease $\sum_{j \in J} 1/b_j$. Hence

$$\sum_{j \in J} \frac{1}{b_j} \geq H(d, \ell). \quad (37)$$

The corresponding allocation value is at most $(\ell - k)/H(d, \ell) \leq \Psi(d, k)$. Conversely, take $m = \ell$, use the balanced composition attaining $H(d, \ell)$, and activate every block. This proves that the worst fixed-composition value, optimized over all compositions, is exactly $\Psi(d, k)$.

D Circuit Covariance Identity

Let $U \in \mathbb{R}^{(b+1) \times b}$ satisfy $U^\top U = I_b$, and let $s \in \ker(U^\top)$ have $\|s\|^2 = a > 0$. Then

$$Q = [U \quad s/\sqrt{a}] \quad (38)$$

is an orthogonal matrix. For omission i , let U_{-i} and s_{-i} remove row i , and put $z = s/\sqrt{a}$. Cofactor identities give the determinant formula below. Orthogonality of row i with every remaining row gives $U_{-i} u_i + z_{-i} z_i = 0$, and therefore

$$\det(U_{-i})^2 = \frac{s_i^2}{a}, \quad U_{-i}^{-1} s_{-i} = -\sqrt{a} \frac{u_i}{s_i/\sqrt{a}}, \quad (39)$$

where $u_i^\top$ is row i of U . Consequently volume sampling has probabilities s_i^2/a and

$$\mathbb{E}[U_{-i}^{-1} s_{-i}] = 0, \quad (40)$$

$$\mathbb{E}[(U_{-i}^{-1} s_{-i})(U_{-i}^{-1} s_{-i})^\top] = a \sum_i u_i u_i^\top = a I_b. \quad (41)$$

The mean identity is also the standard unbiasedness of size- b volume sampling. Mapping intrinsic coordinates back to the range of H_j proves (28). The case $a = 0$ follows by continuity and has zero excess.